4.5 Formation of thyroid hormones

Thyroid gland, a bilobed organ weighing about 20 – 25 grams, is responsible for the synthesis of thyroid hormones. Thyroid hormones are responsible for maintenance of basal metabolic rate. Tri-iodothyronine and tetraiodothyronine are the two important thyroid hormones (Fig 4.15).

The steps involved in the synthesis of thyroid hormones are:

1. Conversion of inorganic iodine to iodide (activation of iodine).
2. Iodination of tyrosine rings in the protein thyroglobulin to form di-iodotyrosine
3. Formation of di-iodothyronine and tetraiodothyronine by coupling and removal of alanine. Thyroglobulin bearing iodinated residues is stored in the thyroid gland.
4. Depending upon the needs of the body, TSH signals the thyroid gland resulting in the proteolysis of thyroglobulin and release of T3 and T4.

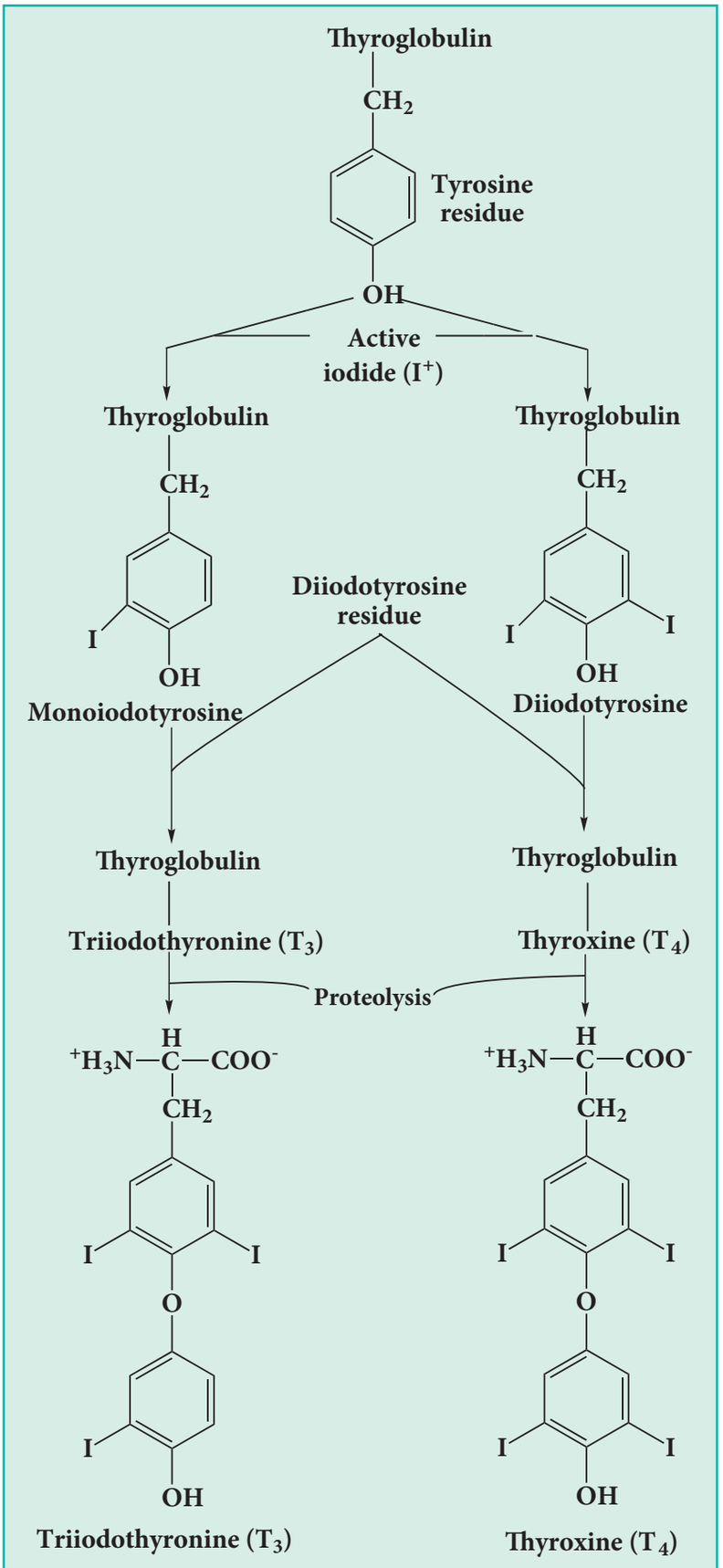

Fig 4.15. Synthesis of thyroid hormones

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